ECHO to Veritas - Master Handoff

Updated: 2026-11-11. Local branch: `main`, based on `9d761e6`. FreeLLMAPI is ECHO's verified primary conversational provider; deterministic commands, security gates and offline fallback are preserved; optional free browser speech is implemented. Phase 1 (conversational follow-up navigation), Phase 4 (Lost & Found) and Phase 5 (attendance, assignments, contacts through a shared tool layer) are complete and verified end to end. ECHO is still routing without model tool-calling: data answers are composed server-side, not generated.

Phase 1 - conversational follow-up navigation (2026-11-09)

The reported defect: after ECHO explained a page and offered to open it, an acceptance such as "okay take me there" only produced more prose. Returning text is not navigation, so the flow was rebuilt around a structured action.

Implemented:

1. **Structured offer lifecycle (server).** An offer is stored per ECHO session with its action ID, path and label, bound to the account `sub` and the role, and expiring after 60 s. `server/services/echo.ts` owns the store; revocation (logout / New chat) deletes it.
2. **Deterministic acceptance before the provider.** Cancellation, explicit replacement ("actually, open assignments instead") and deictic acceptance ("take me there", "open it", "yes please", "show me", "go there", "let's do that", "alright") are resolved before any FreeLLMAPI call, so the primary brain cannot answer them as fresh conversation.
3. **Only offered pages become pending.** The pending action is registered only when the displayed reply names exactly one allowlisted page (or uses the canonical "say open X and I'll take you there" offer). Two destinations yield a clarification, never a guess. Offers can only come from the fixed topic allowlist, never from model-generated URLs.
4. **Verification, not claims.** The `done` SSE event now carries `action: { type: 'navigate', actionId, path, label }`. The widget forwards it via `echo-widget:action`; `EchoAssistant` re-validates the path against `ECHO_PAGES` and the current role, moves the real router, and confirms only after both the React location and `window.location` match. Failures say so.
5. **Client deferral.** `EchoCommandSession` now exposes `hasPending`, and bare follow-ups bypass local recognition (which previously answered "could not match") whenever no local confirmation is active.
6. **Invalidation.** Offers expire after 60 s, are cleared by cancellation, by a different destination, by executing, by an unrelated reply, by a role change, by a second account, and by logout/reset.

Phase 2 - expandable and full-screen workspace (2026-11-09)

The widget panel now has three presentation modes on one mounted conversation: compact (380 px), expanded (min(720 px, 92vw)) and full screen (100vw x 100dvh with a centred readable column). The header gains a size button that cycles compact -> expanded -> full -> compact, with `aria-label` updated per mode ("Expand chat", "Full screen", "Restore size"). Escape restores compact before closing. The mode persists in `sessionStorage` (`echo-widget-mode-v1`) but the

conversation, draft text, speech settings and pending server offers are never unmounted or reset by a mode change.

Verified in the browser regression at 1280x850: the panel widens on expand, fills the viewport in full screen, returns to compact, the URL never changes, and the message count is identical before and after each switch.

Phase 3 - optional voice input and the "Echo" wake word (2026-11-09)

Voice input is opt-in and free: no API key, no paid service, and no audio is sent to the ECHO server. The widget uses the browser's own `SpeechRecognition` (`webkitSpeechRecognition`) engine behind one small provider interface, so a host page can substitute another engine with `window.EchoWidgetRecognitionProviders` and `data-recognition-provider`.

- **Dictation.** The microphone button captures one utterance: interim text appears in the status line, and the final transcript fills the composer and is submitted through the same form path as typing. Portal commands, the FreeLLMAPI conversation, Stop and spoken replies therefore behave identically, and any draft the user already typed is preserved.
- **Wake word.** The "Echo" checkbox arms continuous recognition and acts only on utterances beginning with "Echo", "hey Echo", "okay Echo" or "hi Echo". Any other speech is ignored and never displayed, so room conversation cannot trigger navigation. The wake word is never restored across page loads: every reload needs a fresh opt-in.
- **Release points.** The recognizer is aborted while ECHO speaks (otherwise it transcribes its own voice) and re-armed when the speech queue drains, and it is released on Stop, New chat, logout/widget destroy, panel close and whenever the tab is hidden. Denied permission, a missing device, offline recognition and unsupported browsers produce explicit messages; text chat keeps working and the controls are hidden when no recognizer exists.

Limitations, stated plainly: the browser engine normally transcribes in the vendor's cloud, so audio leaves the device and recognition needs a network; it is missing in some browsers (notably Firefox) and needs microphone permission; and vendors suspend the stream in hidden tabs, so "Echo" is a foreground wake word only, not a background hotword. A true background wake word needs a native app or an always-on local model, neither of which exists here.

Verified with a simulated recognizer in `tests/echo-browser.mjs`: dictation start, interim text, spoken command navigation, one-shot stop, permission denial, Stop while listening, wake-word arming, ignoring speech without the wake word, wake-word command execution, no room-speech display, TTS pause/resume, New chat release, disarm, and a browser without any recognizer. These are simulated-API tests, **not** real microphone or audio verification; no real-device audio run was possible in this environment.

Workspace visualisations remain **not** implemented; Lost & Found retrieval is now implemented as Phase 4 below.

Phase 4 - authorized read-only data, Lost & Found (2026-11-11)

The reported conversation, reproduced and fixed:

> Can you read lost and found and answer me what all are actually found

It previously answered "Communication is for requesting a conversation with a teacher or coordinator...". Cause: `lost`/`found` were keywords of the communication topic, and no tool could reach records at all. Two defects were fixed - the topic mapping, so "open lost and found" no longer offers `/communication`, and the missing data path.

1. **A reader, not a prompt.** `createEchoServiceRouter({ readLostFound })` receives a callback that `server/index.ts` supplies only after the request is authenticated. It runs inside the same account-bound ECHO session, so no role claim or user id ever travels in the message text.
2. **Field allowlist.** `shapeLostFoundItems` in `server/services/echoData.ts` keeps id, type, item name, category, location, date, status and a truncated description. Reporter ids, contact info, claim notes and verification notes are dropped before display, memory or logging.
3. **Grounded answers.** The reply is composed on the server from those records: real counts, names, locations and claim status, plus explicit "nothing matches" and "records unavailable" paths. Lost & Found text is never sent to FreeLLMAPI, and history stores only a summary, so a later provider turn cannot receive the records either.
4. **Working memory.** The last result set is kept per ECHO session (30-minute TTL, deleted on revocation), so "the second one", "claim the blazer" and "which ones haven't been claimed" resolve without re-asking. A second session sees nothing.
5. **Cards.** `done.records` carries `{ id, title, subtitle, meta[], status }` and the widget renders each field with `textContent` only. A record containing markup or an instruction stays inert text.
6. **Stated limits.** Lost & Found is campus-wide and visible to any signed-in account, matching the existing `/api/lost-found` page. Attendance, marks and other per-student records are still not readable through ECHO.

Verified by 12 tests in `tests/echo-lostfound.test.ts`: the exact reported request, a plain list, repeat-turn variation, the unclaimed filter, an ordinal follow-up, a claim follow-up, a no-match search, a missing reader, cross-account isolation, field redaction, prompt-injection record content, and navigation still working. The browser regression also asserts real-database retrieval through the built-in engine, follow-up resolution, redaction in the raw SSE, card rendering and a hostile record that must not execute.

Phase 5 - application-wide module tools (2026-11-11)

The Lost & Found reader is now one instance of a shared layer rather than a special case. `server/services/echoTools.ts` owns shaping, intent detection, aggregation, phrasing and cross-module memory; `server/index.ts` owns scoping.

- **One snapshot, scoped server-side.** `readPortal(identity)` resolves the account id from the ECHO session, looks up that user's student row, and returns only what the existing page APIs would return: contacts (directory-wide), attendance and assignments for that student's own id, class and section. A

staff account gets the directory and nothing else, because a teacher's view spans every student and per-record staff scoping is not designed.
- **Attendance** answers overall rate and per-subject rates from real rows, stating its arithmetic out loud (late counts as attended, excused sessions are excluded). "Which subject is lowest?" is computed, not assumed.
- **Assignments** reads real due dates and submission status: open work, due today/tomorrow/this week, overdue, and per-subject filters.
- **Contacts** searches the published directory by name, role, department and category, and falls back to academic staff for questions about submissions or marks instead of inventing a person.
- **Cross-module memory.** Analysing attendance stores the weakest subject, so "what assignments do I have in that subject?" resolves it. Memory is per ECHO session, 30-minute TTL, cleared on revocation, never shared across accounts.
- **Unavailable is stated.** If the host app cannot read a module, ECHO says so and emits no cards; it never estimates a percentage or a deadline.
- **Cards.** The same `done.records` channel renders module results, text-only.

Module coverage matrix

Module	Tool	Authorization	Status
Attendance	own sessions, per-subject rates	`students.userId = account id`	Implemented, tested
Assignments	own class/section work + own submissions, due/overdue filters	own class & section	Implemented, tested
Contacts	published staff directory	any signed-in account (mirrors `/api/contacts`)	Implemented, tested
Lost & Found	items and claim status	any signed-in account (mirrors `/api/lost-found`)	Implemented, tested
Examinations / marks	none	would need a provider-privacy decision	Not implemented
Timetable	none	read-only source exists (`/api/timetable`)	Not implemented
Announcements	none	read-only source exists (`/api/announcements`)	Not implemented
Leave, library, transport, portfolio	none	—	Not implemented
Staff attendance / assignments	none	per-record staff scoping undesigned	Not implemented

14 tests in `tests/echo-tools.test.ts` cover attendance arithmetic, weakest-subject calculation, cross-module subject follow-up, deadline and overdue filtering, staff search, the academic fallback, an unmatched search, an unreadable module, cross-account isolation, injection inside an assignment title, navigation still winning, and Lost & Found still working beside the new tools. Two intent boundaries were fixed during this work and are asserted by the existing suites: location questions ("where are my marks?") keep the navigation offer, and explanations ("tell me about assignments") are not answered with records.

What changed

1. **FreeLLMAPI is the primary brain.** Natural conversation goes to the FreeLLMAPI OpenAI-compatible gateway. `auto:reliable` is the default model and the gateway chooses the upstream route dynamically. OpenAI is ignored unless `ECHO_ALLOW_PAID_PROVIDER=true`; there is no mandatory paid provider.
2. **Security hardening.** ECHO sessions require a signed-in portal identity, are bound to the account `sub`, reject unapproved origins, use no shipped fallback secret, validate token format/expiry strictly, allowlist only `message` and `page`, bound all stores and rate-limit messages.
3. **Production security hardening (this release):**

- CORS origin allowlisting replaced reflected `origin: true` with explicit
allowlist (403 on foreign origins, same-origin/preflight supported).
- Session secrets enforced: `SESSION_SECRET` ≥32 chars in production;
`assertSessionSecretConfigured()` fails at startup if unset.
- `PORTAL_BOOTSTRAP_PASSWORD` required (≥16 chars) in production; seeds the
principal's password; sample accounts receive random unusable hashes.
- Demo accounts (`/api/auth/demo-accounts`) return 404 in production; generic
forgot-password response instead of password text.
- Production password policy: 12-char minimum for self-service changes;
temporary one-time passwords for admin-created students/staff and resets.
- Login page: dev-only demo account panel; production shows "Secure sign■in
— scrypt-hashed credentials and role■based access control."
- AppShell: dev-only instant role switcher hidden in production.
- StudentsStaffPage: shows temporary password in production enrollment/reset
toasts.
- `server/utils/portalMode.ts`: production-mode detection.
- `server/utils/originPolicy.ts`: new CORS origin allowlist utility.
- `tests/portal-security.test.ts`: 6 new release-gate security tests.
3. **Natural conversation.** Greetings, typos, follow-ups, page/role context,
empathy, jokes, sensitive refusals, provider status and honest no-memory
behavior are handled without inventing school records.
4. **Fuzzy commands.** Politeness, location questions, "pull up", "bring up",
"where is" and location-aware navigation resolve locally. Compound plans are
atomic and execute nothing if any part is unsupported. Ordinary prose with
"and"/"or" stays conversational.
5. **Speech.** Free browser `speechSynthesis`, on/off, Stop, 0.8-1.5x speed,
sentence streaming, no overlap, session-scoped settings and text fallback. A
replacement provider can be registered at `window.EchoWidgetSpeechProviders`.
6. **Provider diagnostics.** Authenticated `/api/echo/provider` plus SSE
`event: meta` reveal provider/model/route without keys, prompts or replies.

Verified provider evidence

Command: `ECHO_USE_DESKTOP_SETTINGS=true node tests/echo-freellm-provider.mjs`.

- Real local gateway, 292 models visible.
- Real multi-turn conversation; synthetic code `QX-7319` recalled on turn 2.
- Provider `freellmapi`, model `command-a-plus-05-2026`, route
`cohere/command-a-plus-05-2026`; turns 1554 ms and 1580 ms.
- The live browser widget received the model-generated marker `PLUMAGE`.
- An earlier run routed to `qwen/qwen3.8-27b` via `groq`, confirming dynamic
routing. Do not hardcode an upstream model in tests.
- Credentials were read from the existing FreeLLMAPI desktop store only with
explicit opt-in, passed to the child server environment, never printed or
written to the repository or browser.

Verified local checks

Check	Result
175 unit/integration tests: config, commands, conversation, navigation, Lost & Found, module tools, security

PASS |
`tsc --noEmit`	PASS
production `vite build`	PASS
full browser regression	PASS
live FreeLLMAPI provider test	PASS (previous release; not rerun here)

`tests/echo-navigation.test.ts` (new) drives the real router and asserts the `done.action` lifecycle: offer-then-accept, `open it`, `yes please`, `show me`, no-pending ambiguity, cancellation, 60 s expiry, explicit replacement, role change, session isolation, logout revocation, cross-account rejection and provider bypass. `tests/echo-browser.mjs` adds a real built-in-engine browser run (real widget, cookie session, real router) proving the URL changes from `/assignments` to `/attendance` only after acceptance, plus a forced `history.pushState` mismatch that must report "could not confirm".

Browser coverage includes login, typo commands, verified navigation feedback, context help, clipboard success/failure, Stop/reset/error recovery, speech on/off/speed/Stop/overlap/failure fallback, voice input (dictation, permission denial, Stop while listening, wake word, TTS pause/resume, unsupported browser), tool record cards, display modes, 390 px mobile, logout abort and second-account isolation. It also drives the real built-in engine against the seeded database: the reported Lost & Found question returns real items, "the second one" resolves, and the raw SSE contains no reporter or contact fields. Browser regression uses fake SSE for command flows and a simulated recognizer; only the opt-in test proves a real model, and no test exercises a real microphone.

Configuration contract

Server-only variables (never `VITE_*`, never frontend code):

```text
FREELLMAPI_BASE_URL=https://<public-https-gateway>/v1
FREELLMAPI_API_KEY=<secret>
FREELLMAPI_MODEL=auto:reliable
ECHO_ALLOW_PAID_PROVIDER=false
ECHO_SESSION_SECRET=<long random value>
SESSION_SECRET=<long random value>
PORTAL_BOOTSTRAP_PASSWORD=<16+ char production bootstrap secret>
```

- Development: with a key and no base URL, ECHO uses `http://127.0.0.1:31415/v1`; the FreeLLMAPI desktop app must be running.
- Production: explicit HTTPS only. Credentials in URLs, query strings, non-loopback HTTP and paths other than `/v1` are rejected.
- `PORTAL_BOOTSTRAP_PASSWORD` is required (≥16 chars) when `NODE_ENV=production` or running on Vercel; seeds the principal's password; sample accounts get random unusable hashes.
- Legacy `FREELMAPI_*` and `ECHO_MODEL_*` aliases still work.
- `/api/echo/provider` reports configured/reachable/model/base host; it never exposes the key.

Vercel status and required action

The deployed site still runs the old build. A model-only prompt received the old scripted welcome, so production is not provider-connected. Vercel cannot reach this PC's loopback gateway at `127.0.0.1:31415`.

To enable FreeLLMAPI in production:

1. Expose the gateway at a public HTTPS origin (an HTTPS tunnel or a hosted FreeLLMAPI-compatible gateway).
2. Add `FREELLMAPI_BASE_URL`, `FREELLMAPI_API_KEY`, `FREELLMAPI_MODEL`, `ECHO_SESSION_SECRET` and `SESSION_SECRET` in Vercel Project Settings -> Environment Variables.
3. Redeploy and sign in.
4. Check `/api/echo/provider` for `configured: true`, `reachable: true`.
5. Send a model-only prompt and confirm the SSE `meta` provider/model.
6. If the values are absent, the deployment intentionally uses deterministic fallback. HTTP 200 or Ready alone is not provider proof.

No credentials were exposed to frontend code. No paid provider is enabled.

Security status

`tests/echo-security.test.ts`: 135/135 PASS (including 6 portal-security gate tests). Fixed issues: anonymous issuance, unapproved origins, usable public fallback secret, malformed expiry, token suffix, unsupported fields, account binding, page-context injection, unbounded state and rate limiting.

New in this release:

- CORS origin allowlisting replaced reflected `origin: true` with explicit allowlist (403 on foreign origins, same-origin/preflight supported).
- Session secrets: `SESSION_SECRET` must contain at least 32 characters in production; `assertSessionSecretConfigured()` fails at startup if unset. `ECHO_SESSION_SECRET` added as alias.
- `PORTAL_BOOTSTRAP_PASSWORD` is required in production (`≥16 chars`); seeds random unusable hashes for sample accounts; only the bootstrap principal logs in.
- Demo accounts (`/api/auth/demo-accounts`) return 404 in production; generic forgot-password response instead of password text.
- Production password policy: 12-character minimum for self-service changes; temporary one-time passwords for admin-created students/staff and password resets.
- Login page: dev-only demo account panel; production shows "Secure sign■in — scrypt-hashed credentials and role■based access control."
- AppShell: dev-only instant role switcher hidden in production.
- StudentsStaffPage: shows temporary password in production enrollment/reset toasts.
- `server/utils/portalMode.ts`: production-mode detection (`NODE_ENV=production` or `VERCEL`).
- `server/utils/originPolicy.ts`: new CORS origin allowlist utility.
- `tests/portal-security.test.ts`: 6 new release-gate security tests.

Residual, pre-existing or architectural:

- `server/utils/auth.ts` still uses a public default cookie secret when `SESSION_SECRET` is unset. Set it privately in production; it is the outer gate.
- Portal-wide reflected CORS remains at the API gateway level; ECHO origin checks are an additional gate, not a replacement.
- History/revocation are process-local; serverless instances do not share them.

- Stop cannot guarantee upstream cancellation/billing.
- Route confirmation covers view state, not lazy page-data loading.
- No physical-device or multi-browser matrix yet.

Key files

- `server/services/echo.ts`: authenticated router, account-bound sessions, origin/body/rate checks, SSE, page context.
- `server/services/echoConversation.ts`: natural conversation, provider priority, FreeLLMAPI/OpenAI config, status/probe, usage metadata, topic allowlist (including the separate Lost & Found topic).
- `server/services/echoData.ts`: authorized read-only Lost & Found tool, field allowlist, intent detection, grounded answer composition, record cards.
- `server/services/echoTools.ts`: shared module tools (attendance, assignments, contacts), intent boundaries, aggregation, cross-module memory.
- `tests/echo-tools.test.ts`: 14 module-tool, memory and authorization tests.
- `tests/echo-lostfound.test.ts`: 12 Lost & Found retrieval, memory and authorization tests.
- `server/index.ts`: authenticated `/api/echo/config`, `/api/echo/provider`, ECHO mount with cookie/Bearer portal identity.
- `src/lib/echoCommands.ts`, `src/lib/echoRecognition.ts`: registry, fuzzy commands, atomic plans, prose guards.
- `echo-web/widget/widget.js`: speech controller, credentials, page context, Stop/reset/copy.
- `tests/echo-navigation.test.ts`: 14 Phase 1 navigation lifecycle tests.
- `src/components/EchoAssistant.tsx`: `echo-widget:action` handler (allowlist, role check, router execution, observed-route confirmation).
- `tests/echo-security.test.ts`: 12 release gates.
- `tests/portal-security.test.ts`: 6 new release-gate security tests.
- `tests/echo-conversation.test.ts`: 26 conversational/provider tests.
- `tests/echo-commands.test.ts`: 85 command tests.
- `tests/echo-browser.mjs`: full browser regression including speech.
- `tests/echo-freellm-provider.mjs`: opt-in real-provider verification.
- `server/utils/originPolicy.ts`: new CORS origin allowlist utility.
- `server/utils/portalMode.ts`: production-mode detection.
- `.env.example`, `README.md`, `echo-web/README.md`: configuration contract.

Git and publication state

Previous release commit: `a49b9db8cacbb9c81490da4b209931ea45a1fa7`. Base for this work: `9d761e6`, rebased onto upstream `5cfd8da` (feat(portal): support PORTAL_DEMO_MODE) before pushing, so history is linear. Phase 1 navigation: `0013123` (feat(echo): execute offered navigation from conversational follow-ups). Phase 2 workspace modes: `6cef2ab` (feat(echo): add expanded and full-screen workspace modes). Phase 3 voice input: `89202cb` (feat(echo): add optional microphone input and a foreground "Echo" wake word). Phase 4 Lost & Found: `708fe80` (feat(echo): answer Lost & Found questions from authorized records). Phase 5 module tools: `4a99067` (feat(echo): generalize authorized read-only tools across modules). Documentation commits sit on top; use `git log --oneline` for the exact current HEAD.

`npm run test:echo`: **175/175 PASS**.
`tsc --noEmit`: PASS.
`npm run build`: PASS (2045 modules).
`npm run test:echo:browser`: PASS, including the real built-in engine run.

Vercel did not auto-deploy the push. The workspace has no `.vercel` project link
or deploy hook. Manual Vercel publish or Git-integration reconnect is required.
After deploying, set environment variables (`SESSION_SECRET`, `ECHO_SESSION_SECRET`,
`PORTAL_BOOTSTRAP_PASSWORD`, `FREELLMAPI_BASE_URL`, `FREELLMAPI_API_KEY`,
`FREELLMAPI_MODEL`, `ECHO_ALLOW_PAID_PROVIDER`) in Vercel Project Settings ->
Environment Variables and redeploy. Do not use `VITE_*` variables. After deploy,
verify `/api/echo/provider` reports `configured: true` and `reachable: true`,
and confirm a model-only prompt returns provider metadata. HTTP 200 or Ready
status alone is not provider proof. Do not force-push.

Remaining priorities (not implemented)

1. **Remaining modules.** Attendance, assignments, contacts and Lost & Found have
tools. Still unsupported through ECHO: examinations/marks, timetable,
announcements, leave, library, transport, portfolio, and any staff-scoped
records. Examinations need a privacy decision before implementation (nothing
private is sent to a provider today; answers would be composed server-side).
Timetable and announcements are the cheapest next additions - both are
read-only and already served by `/api/timetable` and `/api/announcements`.
Staff attendance/assignment rows stay out until per-record staff scoping is
designed, because a teacher's view spans every student.
2. **Charts/tables in the workspace.** `recharts` is already a dependency, but
ECHO renders text only. Structured results need server-validated payloads first,
and only where real supported data exists.
3. **File upload / multimodal input.** Not present.
4. **Real audio and real-provider verification.** Voice input is verified with a
simulated recognizer only. A real microphone run needs a machine with a device
and a foreground browser, which this environment does not provide.

Exact handoff paths

- Markdown:
`F:\Downloads\Veritas-Echo-Integration\echo-web\docs\ECHO-WEB-MASTER-HANDOFF.md`
- PDF:
`F:\Downloads\Veritas-Echo-Integration\echo-web\docs\ECHO-WEB-MASTER-HANDOFF.pdf`
- Backups: `F:\ECHO-Handoff\ECHO-WEB-MASTER-HANDOFF.md` and `.pdf`
- PDF generator: `python echo-web/docs/render_handoff.py` (requires reportlab;
output text is selectable/copyable).

Ready-to-copy continuation prompt

Continue ECHO/Veritas at `F:\Downloads\Veritas-Echo-Integration` on local `main`.
Read this handoff, `PENDING_WORK.md`, `RELEASE_VERIFICATION.md` and
`FUZZY_COMMANDS.md`, then inspect `git status`, `git log` and remotes. Phase 1
(conversational follow-up navigation with a verified, allowlisted route action)
Phase 2 (compact/expanded/full-screen workspace) and Phase 3 (optional browser
microphone dictation plus a foreground-only "Echo" wake word, wired to the same

command/conversation/Stop/TTS path) are complete. FreeLLMAPI is still the primary conversational brain, with deterministic commands, security gates, offline fallback and free browser speech in both directions.

Phase 4 (Lost & Found) and Phase 5 (attendance, assignments and contacts through the shared `server/services/echoTools.ts` layer, with cross-module subject memory) are complete. Next priorities, in order: (1) timetable and announcement tools, then examinations only after a privacy decision; (2) charts for validated numeric results. The portal has no documented grade-projection feature, so do not invent one. Real microphone and real-device audio verification still needs hardware this environment lacks.

Do not recreate completed features, weaken tests or enable a paid provider. After any change, rerun `npm run test:echo` (175 tests), `npx tsc --noEmit`, `npm run build` and `npm run test:echo:browser`; the opt-in provider test needs the FreeLLMAPI desktop gateway. Keep the Markdown/PDF handoffs and `F:\ECHO-Handoff` copies current. Verify the deployed site after pushing; do not claim provider success from HTTP 200 or Ready alone.